30. The heart beat of a person whose head was injured by an accident became irregular and respiration occurred without a control. He happened to vomit also. Thus, it can be inferred that in his brain,
(1) cerebellum is damaged.
(2) medulla oblongata (brain stem) is damaged.
(3) left cerebral hemisphere is damaged.
(4) right cerebral hemisphere is damaged.

31. The magnetic train operated in Japan is lifted a bit up above the rails after it acquires an adequate speed and made to run forward. What is the main advantage obtained by this modern technological strategy?
(1) minimizing train maintenance activities
(2) minimizing the wear of the rails
(3) minimizing air resistance acting on the train
(4) running the train under minimal friction

32. Two reactions used to prepare the gases H_2 and CO_2 in the laboratory are given below.

$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$ $\quad$ $CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$

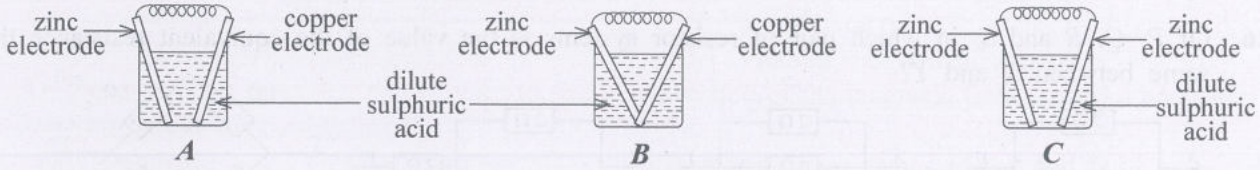
What is the mass of $CaCO_3$ that should be reacted to obtain an amount of moles of CO_2 gas which is equal to the maximum amount of moles of H_2 gas obtainable from 32g of zinc (Zn)? (H = 1; C = 12; O = 16; Ca = 40; Zn = 64)
(1) 22 g (2) 44 g (3) 50 g (4) 100 g

33. In the form of a thin layer, metal *X* is applied on the iron sheet *A* and metal *Y* is applied on the iron sheet *B*. A small scratch was made at the centre of both sheets. Later, rusting has spread rapidly on *A*, but *B* has not been subjected to a considerable damage. The metals *X* and *Y* respectively can be,
(1) Sn and Zn. (2) Zn and Sn. (3) Cu and Ag. (4) Ag and Cu.

34. 600 g of water at temperature 80°C were added to a vessel containing 200 g of water at 0°C. What is the final temperature of the mixture? (specific heat capacity of water = 4200 J $kg^{-1}°C^{-1}$; assume that the vessel does not absorb heat.)
(1) 53°C (2) 60°C (3) 66°C (4) 70°C

35. The areas of cross section of the pistons of a hydraulic jack are x m^2 and $4x$ m^2. How much is the force exerted on the larger piston when a force of 50 N is applied on the smaller piston?
(1) 50 N (2) 100 N (3) 150 N (4) 200 N

36. ***A***, ***B*** and ***C*** are three sets of simple cells.



In the external circuits of the cells ***A***, ***B*** and ***C*** above, an electric current flows only in,
(1) ***A***. (2) ***B***. (3) ***A*** and ***B***. (4) ***A*** and ***C***.

37. Which of the following is a disease which **cannot** be either controlled or prevented by changing the life style?
(1) diabetes
(2) gastritis
(3) chronic kidney disease
(4) haemophilia

38. In which phase of the growth curve of population does the number of organisms in the population reach its carrying capacity?
(1) first phase (2) second phase (3) third phase (4) fourth phase

39. Which of the following statements is true about the ecological pyramids?
(1) Number pyramids are always upright.
(2) In a biomass pyramid, omnivores can exist only as secondary consumers.
(3) Heavy metals accumulate most in the highest level of a biomass pyramid.
(4) Biomagnification is the incorporation of a higher amount of energy in the higher trophic levels of an energy pyramid.

40. As a student says, nitrogenous nutrients in a soil decrease after a flood due to following reasons.
A - occurrence of denitrification
B - percolation deeper into soil (leaching)
C - getting washed away from the surface soil
D - decomposition of organic matter
Of these only,
(1) *A*, *B* and *C* are true.
(2) *A*, *B* and *D* are true.
(3) *A*, *C* and *D* are true.
(4) *B*, *C* and *D* are true.

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